

NEELAY RANJAN

Generative Modeling for Safety-Critical, Data-Scarce Domains

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Generative-modeling researcher building diffusion and language-model systems for safety-critical domains where data is scarce and failure is expensive. First-author work on label-efficient vessel segmentation beats SAM with 7 to 22 labels, and work at NASA Ames covers hazard-aware routing and air traffic control speech pipelines.

RESEARCH & WORK EXPERIENCE

NASA Ames Research Center, Aeronautics Directorate Moffett Field / Mountain View, CA
GenAI Applied Research Intern, SHIFT Aug 2026 – Present

- Fine-tuned Qwen3-Omni-30B with QLoRA for air traffic control (ATC) speech-to-text on ~1.5 h of real audio (~900 utterances), reaching **17% WER vs. 20% for the existing Whisper ASR** and 30% zero-shot in 5-fold CV.
- Replaced LLM parsing of transcripts into the DTI maneuver ontology with Kev, a typed decision model that cannot invent fields. It **raised value-match accuracy from 40% to 80%** over the best rules-based and LLM attempts, at ~90% lower latency and ~400× lower cost than Qwen3 27B.
- Building a text-to-speech synthetic ATC corpus that expands the dataset ~20× (5% real, 95% synthetic). Its outputs feed a maneuver database matched to historical and live FAA airspace records for safety research.

GenAI Applied Research Intern, SLAAC May – Aug 2026

- Built hazard-aware flight rerouting with diffusion models learned from FAA radar tracks. Guidance applied mid-sampling steers around special-use airspace at inference with no retraining, and **99% of reroutes clear the 25 nm hazard buffer at a median +10 nm (1.1%) added distance**, within 1.2% of the geometric optimum.
- Designed a flight-plan LLM with a custom tokenizer that synthesizes a full day of ~**44,000 FAA-managed flights** matched to historical traffic density, producing simulation-ready traffic days for the software NASA and the FAA use to evaluate future air traffic management (ATM) strategies.

Regenstrief Institute Indianapolis, IN
Applied Vision-AI Researcher Feb 2024 – Present

- Showed an x0 flipped-objective diffusion segmenter outperforms SAM (ViT-B) on benchmark angiograms in ultra-low-data regimes, reaching **88.2% Dice with just 16 labels**. Its pre-labels bootstrap surgeon labeling and **cut correction time by 75%**.
- Built a modular SAM-x0 tiled refinement pipeline (SAM prior + partial-resolution x0 diffusion refiner) for real-world, region-agnostic angiograms at native 1024 resolution. It reaches **70% Dice with 22 labeled images**, beating both SAM (46%) and raw x0 (32%) at a fraction of full-resolution compute.
- Extended the pipeline to cerebral moyamoya MRA, segmenting vasculature from **only 7 labeled images** with patient-level cross-validation and tiling augmentation, at ~55% Dice vs. SAM’s ~20%.

SELECTED PUBLICATIONS

- N. Ranjan et al. “Bootstrapping surgeon labeling campaigns with x0-diffusion: label-efficient vessel segmentation of catheter-based angiograms.” *First author, in preparation.*
- N. Ranjan et al. “Modular SAM-prior diffusion refinement for real-world angiogram and seven-shot moyamoya MRA vessel segmentation.” *First author, in preparation.*
- Co-author, “Helical Antenna for Electromagnetic Field Stimulation in Alzheimer’s Disease Therapy.” *Presented at IEEE MWSCAS 2026. Led the PCB design team.*

PROJECTS & APPLIED RESEARCH

Solar-Powered Energy-Based Chess Engine 2025

- Trained an energy-based model that scores and ranks moves from a learned energy landscape over game states, reaching ~**1900–2200 estimated Elo** against a ladder of rated Stockfish depths.
- Embedded the EBM on a Raspberry Pi Zero 2 W (~1 s per move) as a standalone chess engine **powered solely by onboard solar & supercapacitors**. Quantization tests showed FP32 beat INT8 on ARM, which had no native INT8 kernels.

LLM Design Team Lead, V2X Aircraft Maintenance Assistant Aug 2024 – May 2025

- Led design and fine-tuning of a maintenance-assistant LLM for V2X, a Raytheon divestiture. Its two-stage RAG pipeline picks sources from their summaries, then pinpoints and quotes the relevant section, **cutting hallucinations from ~40% to ~5%**.

KEY RESULTS

- 17% WER vs. 20%**
QLoRA ATC speech-to-text, 5-fold CV
- 80% value match vs. 40%**
Kev typed parsing, 90% lower latency
- 99% hazard clearance**
Diffusion reroutes at +1.1% distance
- 88.2% Dice at 16 labels**
x0 segmenter vs. SAM on angiograms
- 75% faster correction**
x0 pre-labels speed surgeon labeling
- 70% Dice at 1024 px**
Real-world angiograms (SAM 46%)
- 7 training images**
Moyamoya MRA at ~55% (SAM ~20%)

V2X **5% hallucinations vs. 40%**
Two-stage RAG, maintenance LLM

EDUCATION

Purdue University, College of Engineering
B.S. in Artificial Intelligence
 Expected May 2027 Indianapolis, IN
Concentration: Intelligent Control & Systems
Minor: Mathematics
GPA: 3.68 / 4.0 · Dean’s List
John Martinson Honors College
CITI: Biomedical Research, RCR, GCP

TECHNICAL SKILLS

- Generative modeling**
foundation-model adaptation (SAM)
diffusion (x0 / v-prediction) transformers
energy-based models LLM fine-tuning & RAG
safety-critical AI interpretability
multimodal pipelines
- ML engineering**
Python C/C++ CUDA PyTorch Git
medical image segmentation Docker W&B
Linux / cluster administration quantization
transfer learning Slurm / H100
- Hardware / edge**
PCB design ultra-low-power inference
embedded ML